

Micro-LED Mass Transfer Technologies

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Abstract—Micro-LED is an emerging optoelectronic technology with the advantages of high brightness, self-illumination, fast response time and long life. It is considered to be the next generation display technology and has attracted the attention of researchers and display companies around the world. But the further development of micro-LED still faces some difficulties, one of which is the mass transfer. A variety of technologies have been used to solve the problem of mass transfer. This article will discuss several main solutions currently being developed.

Keywords—micro-LED, mass transfer

I. INTRODUCTION

Micro-LED is an emerging technology in the field of optoelectronics. It is a miniaturized LED with a size between a few microns and a few hundred microns. It has been developed for 20 years since its introduction. Compared with LCD, it has the advantages of self-luminescence, compared with QLED, it has the advantages of fast response, and compared with OLED, it has the advantages of high brightness and long life. The advantages of micro-LED make it have application prospects in many photoelectric fields. In the display field, micro-LED can be applied to VR/AR screens and watch screens with small area and high pixel density, and can also be applied to large-area display screens. Such as TV screen or outdoor display. In the field of communication, micro-LED can be used for indoor optical communication, underwater optical communication, and has good reliability under the current density of kA/cm^2 level. After decades of development, micro-LED has been favored by many companies and has achieved some commercial results.

For example, at the CES consumer show in 2020, Samsung showed a 292-inch 8K resolution passively driven micro-LED TV display "The Wall", and a "The Window" series of 150-inch 8K resolution active-driven micro-LED

display screen. The left picture in figure 1 shows the "The wall" exhibited by Samsung in 2018.

Ledman currently has 2K, 81-inch, 4K, 162-inch, 8K, 324-inch and other micro-LED display products based on COB microdisplay technology. And in February 2020, a micro-LED ultra-high-definition display with a dot pitch of P0.6 was exhibited. Ledman's micro-LED display products have been successfully used in security, surveillance, education, command centers, business centers, conference centers and other professional fields.

In July 2019, PlayNitride established a micro-LED mass production line to provide micro-LED chips. In 2020, TIANMA and PlayNitride launched the world's first 7.56-inch AM micro-LED display with a transmission rate of more than 60%. The display resolution is 720*480. On behalf of Tianma's development and exploration in the field of new technology of vehicle display, it means that micro-LED can be applied to the head-up display (HUD) of the driver.

And at CES 2020, KONKA exhibited "Smart Wall" with micro-LED technology, including 118-inch 4K and 236-inch 8K micro-LED displays, as shown in figure 1. Using ultra-high-precision mass transfer technology to achieve the transfer and bonding of nearly 100 million micro-LED chips. Through modular splicing, various new display application scenarios and possibilities can be created.

Although micro-LEDs have made achievements in the field of display applications, there is still a certain distance from large-scale commercialization. The problems faced by the commercialization of micro-LED include full color, wavelength consistency, and mass transfer, which limit the production of micro-LED. Mass transfer refers to the transfer of millions of three-color micro-LED devices from a growth substrate to a target substrate to provide electrical connection and mechanical support.



Fig. 1. Samsung "The Wall" shown on CES 2018 (left). KONKA "Smart Wall" shown on CES 2020 (right).

TABLE I. MASS TRANSFER TECHNOLOGIES

Technology	Representative Party	Means	Placement Rate [2]	16K ultra large screen	4K TV	2K cell phone	Car display
Elastomer Stamp (μ TP)	X-Celeprint	van der Waals	1M units/hr	398 h	25 h	11 h	4 h
Roll Printing	KIMM	van der Waals	1000 units/sec	111 h	7h	3 h	1 h
Laser Release	Uniqarta, QMAT	Laser	100M units/hr	4 h	0.25 h	0.11 h	0.04 h
Liquid Assembly	eLux	Gravity & fluid force	50M units/hr	8 h	0.50 h	0.22 h	0.08 h
Magnetic Stamp	ITRI	Magnetic force	0.9M units/hr	442 h	27 h	12 h	5 h
Electrostatic Stamp	Luxvue	Electrostatic force	12M units/hr	33 h	2 h	0.92 h	0.35 h

When the size is small, the micro-LED chip can be integrated into the driving circuit in one piece by methods such as flip chip [1]. However, on displays larger than 2 inches, mass transfers are practical, which can provide micro-LEDs with more flexible electric drive, reduce response time, and can integrate three-color micro-LEDs on a pixel. Mass transfer technology has been researched by multiple companies and university research teams, and has obtained multiple solutions. The solution can be divided into precise pick-up and placement, roll-printing, selective release, and self-assembly according to the transfer methods. Specifically including elastomer stamp micro-transfer-printing (μ TP) [3,4], roll-printing [5], laser release [6], liquid assembly, magnetic stamp transfer, and electrostatic stamp transfer [7]. Table 1 shows the transfer rates of various methods.

Due to different transfer rates, the time required to produce a display screen using various transfer technologies is also different. For ultra large screen with 16K resolution, 398,131,200 three-color pixels need to be integrated on an over 200 inches screen. For 4K resolution, 75-inch TV screens, 24,883,200 pixels need to be integrated. For a 2K resolution, 6-inch mobile phone screen, 11,059,200 three-color pixels need to be integrated. For car on-board displays, such as the 12.1-inch on-board display proposed by AUO, contains 4,147,200 pixels. Therefore, different mass transfer methods are suitable for applications with different needs. Table 1 discusses the time required by various technologies under various resolution requirements.

II. PRECISE PICK-UP AND PLACE

Mass transfer technology based on the idea of precise pick-up and placement includes elastomer stamp micro-transfer-printing, magnetic stamp transfer, and electrostatic stamp transfer. The common feature of these technologies is the use of micro print heads to accurately pick up the corresponding micro-LED devices, and then accurately placed on the target substrate, as shown in figure 2.

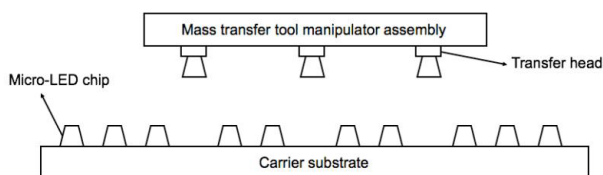


Fig. 2. The common feature of pick-up and placement.

Micro-transfer-printing is a mass transfer technology that has been in development for a long time. The basic idea is to use elastic stamps to pick up, transfer, and print micro-LED devices from the growth substrate to the target substrate, usually a glass substrate or flexible Substrate. It is the viscosity difference between the two interfaces that determines the picking and placing process. The two interfaces include the interface between the stamp and the device, and the interface between the device and the substrate. Based on different ideas for controlling viscosity, micro-transfer-printing has many variants. Including controlling the peeling rate [8], controlling the curvature of stamps [9], making microstructures [10,11], using laser ablation [12], using gas control [13], using shape memory polymers[14]. At present, the placement rate of micro-transfer-printing technology can reach 1M units/hr, but it is not enough for the production of large-area high-resolution display screens. Micro-transfer-printing is more suitable for small-area displays, such as watch screens, VR/AR screens, etc. In addition, since micro-transfer-printing uses flexible stamps made of soft polymers, usually polydimethylsiloxane (PDMS), it can fit any shape of target substrate. Therefore, micro-transfer-printing is more suitable for making wearable micro-LED devices than other mass transfer technologies.

Magnetic stamp transfer and electrostatic transfer are similar to micro-transfer-printing. Magnetic stamp transfer mixes magnetic materials into the manufacturing materials of micro-LEDs, uses coils as transfer heads, and uses the electromagnetic force generated when energized to attract and place micro-LED devices. Electrostatic transfer technology makes an electrode structure on the transfer head, and generates electrostatic force by applying a voltage to attract and clamp the micro-LED device. These two technologies are also not fast, but have the advantage of selective transfer.

III. SELF ASSEMBLY

Self-assembly technology refers to making a large number of three-color micro-LEDs automatically move to a reserved position when passing through the substrate by making a substrate structure or a magnetic structure. This method does not require the use of optical alignment. At present, there are two main methods, one is represented by eLux, which is called fluid self-assembly. During the assembly process, the electrodes are mechanically and electrically connected through a liquid medium to capture and calibrate the micro-LEDs on the solder joints, as shown in figure 3 [7]. Another method, represented by SelfArray, uses magnetically-oriented directional self-assembly. First, the LED appearance is coated

with a layer of pyrolytic graphite film, which is placed on a vibrating magnetic platform. Under the guidance of the magnetic field, the LED will be quickly aligned to the positioning. Self-assembly technology has the advantages of low cost and high output, and has great potential in the field of large-scale display.

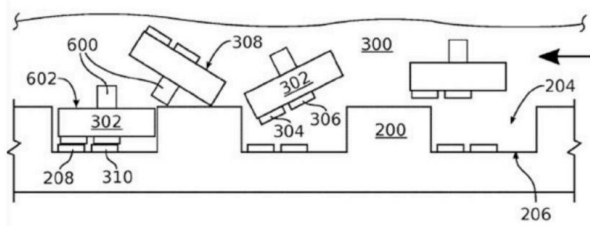


Fig. 3. Schematic diagram of fluid self-assembly.

IV. SELECTIVE RELEASE

Selective release is a method of quickly transferring micro-LEDs. LED devices need to be transferred from the original wafer to a glass substrate with a sacrificial layer. Use a single laser to detect and remove defective LED devices, and then use a UV laser to ablate the sacrificial layer to generate nitrogen gas. Through the 10-300 micron gap created by the gas, the device is transferred to the receiving substrate, and then the missing part is filled with a single laser. This method has a good ability to detect defects, and the ability to transfer at high speed, is expected to achieve rapid production of micro-LED large-area display.

V. CONCLUSION

In conclusion, we reviewed several methods of mass transfer. Mass transfer is an important foundation for the commercialization of micro-LED, and a variety of technologies have been applied to micro-LED mass transfer. Mass transfers still need to be optimized to further develop micro-LEDs.

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